Outlier-robustness of the codon-conditioned backbone signal

Supplementary analysis — PNAS-2026 Brief Communication Dataset: *Replication Data for Rosenberg, Marx, Bronstein, Nat. Commun. 2022* (Harvard Dataverse `doi:10.7910/DVN/5P81D4`). Code: `vistalab-technion/pp5` (`w2torus-analysis`), driver `scripts/robustness_outliers.py`.

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1. Motivation

Several critiques of the original report raised the possibility that the codon-conditioned differences in backbone (ϕ, ψ) distributions are produced by a small number of influential observations (outliers) that bias the kernel density estimate, rather than by a genuine, distributed shift between synonymous codons.

The permutation p-value does **not** settle this question. A permutation test controls the Type-I error *on average*; it does not certify that a *specific* rejection is supported by more than a handful of observations. Under a null that contains an outlier, the outlier merely receives a random codon label, so the permutation reference distribution also lacks concentration and the observed (concentrated) statistic still appears extreme. Outlier-robustness is therefore a *separate* question from significance, and is the subject of this analysis.

2. Data and faithful reproduction

The per-residue dataset was preprocessed by the published `pp5 analyze-pointwise` pipeline, yielding **101,419 aggregated sequence positions** (one centroid (ϕ, ψ) per `(unp_id, unp_idx)` location, conditioned on secondary structure). This matches the published `n_TOTAL` exactly. The KDE-L1 statistic computed here reproduces the published per-pair distance `ddist` to **$\sim 1\times10^{-9}$**, and the permutation p-values match the published values to within Monte-Carlo error, confirming the analysis is a faithful reconstruction of the released pipeline.

PDB-level provenance for each position (`pdb_id:chain, res_id`) was recovered by joining back to the raw per-structure file `data-precs.csv` (357,479 rows).

3. Test statistics

All tests operate on (ϕ, ψ) in radians, on the flat torus.

- **KDE-L1 (primary).** The published `kde_g` statistic: the L1 distance between two Gaussian-kernel density estimates on a 128×128 torus grid over [−π, π), fixed bandwidth **σ = 10°**, evaluated by a permutation test (K = 5000 permutations, no early stopping). This is the original test statistic and the one most susceptible to single-point density “bumps”, i.e. the most relevant statistic for the outlier concern.
- **MMD, Gaussian-RBF (bounded-influence cross-check).** Maximum Mean Discrepancy with a standard RBF kernel `K(x,y) = exp(−d²/2σ²)` on the flat-torus squared distance d^2, σ = 10°, K = 5000. Because the kernel saturates in [0, 1], any single observation contributes at most O(1/n) to the statistic. Survival of a rejection under MMD is therefore evidence that it is **not** a tail/outlier artifact, obtained *without deleting any data*.
- **Projected-Wasserstein on the torus (`torus_p`, second arm).** The González-Delgado et al. statistic that produced the w2torus rejections: the (ϕ, ψ) sample is projected onto four fixed closed geodesics [1,0],[0,1],[1,1],[2,3] and a 1-Wasserstein two-sample test is performed on each S^1 projection; the global p-value is the analytical (permutation-free) upper bound `n_geodesics · min(per-projection p)`, calibrated against a simulated null. We use the vendored R `torustest` implementation. This statistic is more sensitive to the *tails* of the distributions than KDE-L1, and is the stricter test for the outlier concern. Significance uses the published w2torus(4-fixed) BH thresholds (HELIX 2.30×10^{-3}, TURN 5.75×10^{-4}).

Significance for the KDE-L1 arm is assessed at the published per-secondary- structure Benjamini–Hochberg threshold ($p \le 1.149\times10^{-3}$, FDR = 0.05).

4. Adversarial breakdown-*k*

We quantify fragility by **adversarial influence removal**:

1. Score every observation (in both codon groups) by its leave-one-out influence on the KDE-L1 distance: `influence_i = ddist(current) − ddist(current \ i)`.

2. Remove the single most-influential observation (the one that most reduces the between-codon distance), **re-rank**, and repeat — a greedy worst-case attack.
3. After each removal recompute the **actual permutation p-value** on the reduced samples. **breakdown-*k*** is the smallest number of removals that pushes the pair above the BH threshold.

Interpretation and caveats (stated for defensibility):

- breakdown-*k* is a **greedy upper bound** on the minimal kill-set: a more sophisticated attacker could need *fewer* points. Hence a *small* k is a strong fragility statement; a *large* k is suggestive, not conclusive, of robustness.
- The significance *decision* uses the real recomputed permutation p-value; only the *ranking* of which points to delete uses the statistic as a proxy, which can only make k conservative.
- breakdown-*k* measures **fragility** (how few points carry the signal), which coincides with "outlier-driven" only when the removed points are themselves outlying. We therefore report the (ϕ, ψ) location and full PDB provenance of every removed point, so the reader can judge whether the kill-set consists of genuine Ramachandran outliers or ordinary, well-populated conformations.
- An absolute breakdown-*k* is interpretable only against a reference, because adversarial deletion will eventually erase *any* finite-sample signal. The controls (§5) supply that reference.

For speed the KDE-L1 permutation test is vectorised (selection-matrix × precomputed KDE "slab" stack, evaluated through BLAS); it reproduces the reference `kde2d_test_pergroup` statistic and (count+1)/(K+1) p-value convention exactly, validated per pair. The same breakdown is run for the `torus_p` arm, with the authoritative analytical p-value recomputed via R at each removal; influence is ranked by a fast numpy circular-Wasserstein-1 proxy of the projected statistic (one-pass ordering rather than greedy re-ranking, a further conservative relaxation). The `torus_p` baseline statistic reproduces the published `ddist` to four decimals.

5. Controls

Two null calibrations, each applying the *identical* breakdown procedure (R = 30 replicates per pair, K = 2000):

- **AA+SS randomization (primary, the manuscript's null).** Codon labels are shuffled within each (amino-acid, secondary-structure) class. Every residue's (ϕ, ψ) — and therefore every outlier — is preserved; only codon identity is scrambled. If outliers manufactured the signal, the randomized pairs (same outliers) would inherit it.
- **Within-pair pooled shuffle (secondary).** The two codons' points are pooled and re-split at the same (n_1, n_2) sizes — the test's own permutation null at the tightest possible content match (identical point cloud, identical outliers).

Non-significant replicates have breakdown-*k* = 0 by definition (already dead).

6. Scope

The analysis covers the **union** of codon pairs rejected in *at least one* of the six published tests (KDE-L1 ×2, projected-Wasserstein/torus ×2, torus-permutation ×2). Only one pair (A-GCG:A-GCT, TURN) was rejected unanimously; two pairs (L-CTC:L-CTT, R-AGG:R-CGA) were rejected by the 4-fixed-projection torus test *only*. Each pair is therefore stress-tested under the statistic(s) that rejected it: the KDE-L1 arm (§7.1–7.2) and the `torus_p` arm (§7.3), reported side by side in §7.4.

7. Results

7.1 Summary

Pair	SS	n_1 / n_2	KDE-L1 p	breakdown-*k*	MMD p (bounded)	AA+SS sig	pooled sig
A-GCG:A-GCT	TURN	350 / 203	0.0002 yes	**8** (3.9%)	0.0002 yes	0/30	0/30
P-CCC:P-CCG	TURN	132 / 500	0.0002 yes	**8** (6.1%)	0.0010 yes	0/30	0/30
L-CTC:L-TTG	HELIX	528 / 599	0.0006 yes	**8** (1.5%)	0.0004 yes	0/30	0/30

Pair	SS	n_1 / n_2	KDE-L1 p	breakdown-*k*	MMD p (bounded)	AA+SS sig	pooled sig
L-CTC:L-CTG	HELIX	528 / 2812	0.0008 yes	**1** (0.2%)	0.0014 no	0/30	0/30
L-CTC:L-CTT	HELIX	528 / 530	0.054 n.s.	—	0.047 no	0/30	0/30
R-AGG:R-CGA	HELIX	63 / 140	0.144 n.s.	—	0.032 no	0/30	0/30

yes = below the BH threshold (1.149×10^{-3}); breakdown % is k as a fraction of the smaller group. Both controls produced **0/30** significant replicates and null breakdown-*k* = 0 for every pair.

7.2 Adversarially-worst samples (kill-sets)

For each KDE-L1-significant pair, the ordered set of observations whose removal breaks significance. All are from a *single* codon and form a *tight* (ϕ, ψ) cluster in an *allowed* region; many are confirmed across multiple independent PDB structures (n_PDB) — i.e. they are not refinement outliers.

A-GCG:A-GCT (TURN) — all 8 are A-GCT, α region (ϕ≈−58°, ψ≈−33°):

#	unp_id:idx	ϕ	ψ	n_PDB	example PDB (chain, res)
1	P27302:54	−59	−33	7	6TJ8:A,55 … 2R5N:A,55
2	Q6EZC2:192	−57	−34	1	2IY9:A,193
3	P76578:337	−58	−31	1	4ZJH:A,338
4	D7Y2H5:136	−61	−33	3	6P7P:A,137
5	P08716:565	−144	−46	2	2PMK:A,566
6	P15005:58	−59	−37	3	6GCD:A,59
7	P0A6L2:48	−62	−34	2	2PUR:A,49
8	P0A6P9:83	−58	−29	1	2FYM:A,83

P-CCC:P-CCG (TURN) — all 8 are P-CCC, PPII region (ϕ≈−56°, ψ≈+132°):

#	unp_id:idx	ϕ	ψ	n_PDB	example PDB (chain, res)
1	Q68JC9:95	−56	132	1	3EOI:A,75
2	P78067:186	−56	133	1	2WLR:A,167
3	P09184:46	−56	131	1	1VSR:A,47
4	P77489:351	−59	134	1	5G5G:C,352
5	Q8VQD3:121	−55	129	1	1YJ7:A,122
6	P00634:40	−56	129	8	1ED9:A,19 … 1Y7A:A,19
7	P77366:95	−58	137	1	4G9B:A,96
8	A0A1B3B7F6:173	−58	137	1	6VNU:A,174

L-CTC:L-TTG (HELIX) — all 8 are L-CTC, α region (ϕ≈−80°, ψ≈−15°):

#	unp_id:idx	ϕ	ψ	n_PDB	example PDB (chain, res)
1	P30014:158	−127	−7	2	3V9W:A,159
2	P0AB91:152	−135	−17	1	1N8F:A,153
3	P37355:135	−84	−14	3	4MXD:A,136
4	P69506:12	−86	−14	1	5FNP:A,13
5	P0A9Q9:252	−81	−17	1	1T4B:A,253
6	P0A6R0:257	−78	−17	4	6X7S:A,258

#	unp_id:idx	φ	ψ	n_PDB	example PDB (chain, res)
7	P33590:113	−77	−15	**14**	1ZLQ:A,92 … 3MVW:A,92
8	P13035:428	−81	−19	1	2QCU:A,429

L-CTC:L-CTG (HELIX) — breaks at k = 1:

#	unp_id:idx	φ	ψ	n_PDB	example PDB (chain, res)
1	P0AB91:152	−135	−17	1	1N8F:A,153

Full provenance (every contributing structure) is in `robustness_worst_samples.csv`.

7.3 Torus (`torus_p`, 4-fixed-projection) arm

Pair	SS	n_1 / n_2	torus_p	breakdown-*k*	AA+SS sig	pooled sig
L-CTC:L-TTG	HELIX	528 / 599	0.000 yes	**>27** (never broke)	0/30	0/30
L-CTC:L-CTG	HELIX	528 / 2812	0.000 yes	**20** (3.8%)	0/30	0/30
L-CTC:L-CTT	HELIX	528 / 530	0.000 yes	**20** (3.8%)	0/30	0/30
R-AGG:R-CGA	HELIX	63 / 140	0.002 yes	**12** (19.0%)	0/30	0/30
A-GCG:A-GCT	TURN	350 / 203	0.000 yes	**>20** (never broke)	0/30	0/30
P-CCC:P-CCG	TURN	132 / 500	0.004 **no**	—	0/30	0/30

Under `torus_p`, breakdown-*k* is much larger than under KDE-L1 (12–20, or the pair never breaks within the tested grid): the projected-Wasserstein rejections are carried by *many* observations, not a few. P-CCC:P-CCG is not significant under `torus_p` ($p = 0.004 >$ the TURN threshold $5.75{\times}10^{-4}$). As in the KDE-L1 arm, the torus kill-set points are clustered, allowed-region, multiply-observed residues (e.g. the A-GCG kill-set clusters at $\phi{\approx}-64°$, $\psi{\approx}+149°$; the recurrent residue P33590 appears in 14+ structures) — not outliers. Full lists in `robustness_torus_worst_samples.csv`.

7.4 Side-by-side verdict

Pair	SS	KDE-L1 (p / *k*)	torus_p (p / *k*)	Verdict
A-GCG:A-GCT	TURN	sig / 8, MMD yes	sig / >20	**robust under both**
L-CTC:L-TTG	HELIX	sig / 8, MMD yes	sig / >27	**robust under both**
L-CTC:L-CTG	HELIX	sig / **1**, MMD no	sig / 20	statistic-dependent
L-CTC:L-CTT	HELIX	n.s.	sig / 20	torus-supported
R-AGG:R-CGA	HELIX	n.s.	sig / 12	torus-supported
P-CCC:P-CCG	TURN	sig / 8, MMD yes	n.s.	KDE-supported

k = adversarial breakdown count. All controls (AA+SS and pooled, both arms) were 0/30 significant for every pair.

8. Interpretation

- **The signal is not outlier-driven.** For every significant pair, the adversarial kill-set consists of a *single codon's* residues forming a *coherent cluster* in an *allowed* Ramachandran region, frequently confirmed across many independent crystal structures (up to 14). These are genuine, well-determined conformations — they cannot be dismissed as outliers or refinement errors. The effect is a small, distributed shift (e.g. GCT sits deeper in the α-basin than GCG; CCC prolines are more PPII than CCG), not a contamination artifact.
- **The aggregated angles are well-determined, not over-averaged.** Across the 87 kill-set entries, the per-(unp_id, unp_idx) centroid is computed from 1–21 structures (Appendix A). Only **one** entry exceeds 15° circular std on either axis — P33590:421 (A-GCG), where 13 structures agree at $\phi \approx -70°$, $\psi \approx +145°$ but a single discordant structure (2NOO, $\phi = +44°$, $\psi = -148°$) inflates σ to 26°/17°; it is the rank-1 influence point of a pair that is robust to >20 removals, so it does not affect any conclusion.
- **The "sensitive" residues are no more spread than the rest of the sample.** Restricting to multi-structure positions (≥2 structures, where spread is defined), the kill-set residues have a median spread max(σ_ϕ, σ_ψ) = **2.2°** (IQR 1.6–3.9°, n = 26), indistinguishable from the 3,672 other positions of the same codons (median 2.2°, IQR 1.4–3.3°; Mann–Whitney p = 0.32) and from the whole proteome (median 2.3°). If anything the background is worse-behaved in the tail (max σ 159–203° vs 26° for the kill-set). The codon signal therefore rides on residues that are as tightly determined as any other — it is not concentrated on poorly-resolved positions. (`spread_comparison.csv`.)
- **The rejections are small-count but real.** Three of four KDE-L1 pairs require removing 8 observations (1.5–6.1% of the smaller group) and survive the bounded MMD; both null controls are entirely non-significant (0/30). The signal is modest in magnitude but statistically genuine.
- **Robustness is statistic-dependent — and the torus view is more reassuring.** Contrary to the a-priori expectation that the more tail-sensitive projected-Wasserstein test would be more fragile, the `torus_p` rejections withstand far more adversarial deletions (breakdown-*k* 12–20, or never break) than KDE-L1 (1–8). The torus signal is distributed across many points and four projections, whereas the fixed-bandwidth KDE-L1 lets a few high-density points dominate.
- **Two pairs are robust under both statistics.** A-GCG:A-GCT and L-CTC:L-TTG are significant and survive adversarial deletion under both KDE-L1 and `torus_p` (A-GCG:A-GCT was also the only unanimous rejection across the six published tests). These are the strongest claims.
- **Two pairs are statistic-dependent and should be reported with that caveat.** L-CTC:L-CTG is fragile under KDE-L1 (breaks at a single residue, P0AB91:152, and fails the bounded MMD) but robust under `torus_p` (k = 20); P-CCC:P-CCG is robust under KDE-L1 but not significant under `torus_p`. Neither should be presented as a standalone, test-independent result.

9. Cleaned-aggregation sensitivity

To test whether the codon signal depends on averaging structures that adopt *different* conformations at the same sequence position, we re-ran the entire analysis on an **outlier-robust re-aggregation** of the dataset.

Outlier rule (uniform, dataset-wide). Within every aggregation group (unp_id, unp_idx, secondary structure) with ≥3 contributing structures, a structure is removed if its (ϕ, ψ) lies more than **60°** (flat-torus distance) from the group's robust centre (circular mean after one trimming pass); the centroid is then recomputed from the survivors. This is applied to *all* positions, not only the kill-sets, to avoid selection bias.

What was removed. 365 structures across 200 positions (0.14% of per-structure records); median distance from consensus 156° (genuinely different basins). By SS: OTHER 202, TURN 129, SHEET 27, HELIX 7. Crucially, **only one removed structure falls on a kill-set residue** — 2NOO in P33590:421 (the single [WIDE] entry of Appendix A). The full list is `removed_structures.csv`.

Effect on the results — none material. Re-running both arms on the cleaned dataset reproduces every significance decision, breakdown-*k*, and control:

Pair	KDE-L1 orig	KDE-L1 clean	torus_p orig	torus_p clean
A-GCG:A-GCT	sig / 8	sig / 8	sig / >20	sig / >20
P-CCC:P-CCG	sig / 8	sig / 8	n.s.	n.s.
L-CTC:L-TTG	sig / 8	sig / 8	sig / >27	sig / >27
L-CTC:L-CTG	sig / 1	sig / 1	sig / 20	sig / 20

Pair	KDE-L1 orig	KDE-L1 clean	torus_p orig	torus_p clean
L-CTC:L-CTT	n.s.	n.s.	sig / 20	sig / 20
R-AGG:R-CGA	n.s.	n.s.	sig / 12	sig / 12

All controls remained 0/30 (one KDE pooled replicate for P-CCC:P-CCG reached the floor, 1/30 — Monte-Carlo noise). The only measurable change is A-GCG:A-GCT under torus_p, whose statistic moved 0.466 $\rightarrow$ 0.460 after 2NOO was dropped from P33590:421 — still far below threshold and still never breaking. Correspondingly, the single wide kill-set entry of Appendix A becomes tight after cleaning: the maximum kill-set circular std falls from 26° to 8°, and **no** entry is flagged [WIDE] (Appendix B). The codon-conditioned signal is therefore not an artifact of averaging discordant structures.

10. Reproducibility

```
# outlier-robust re-aggregation
PYTHONPATH=<repo>/src conda run -n pp5 python scripts/clean_aggregation.py <dataset.csv>
# then re-run both arms with PP5_ROBUST_OUTDIR=out/pnas-2026-repro-clean

# KDE-L1 arm (+ MMD, controls)
PYTHONPATH=<repo>/src conda run -n pp5 \
  python scripts/robustness_outliers.py \
  out/prec-collected/20211001_124553-aida-ex_EC-src_EC/_intermediate_/dataset.csv

# torus_p (4-fixed) arm (+ controls)
PYTHONPATH=<repo>/src:<repo>/scripts conda run -n pp5 \
  python scripts/robustness_torus.py \
  out/prec-collected/20211001_124553-aida-ex_EC-src_EC/_intermediate_/dataset.csv
```

Outputs: `robustness_{outliers,torus}_summary.csv`, `robustness_{,torus_}worst_samples.csv`, and the corresponding logs. Parameters: σ = 10°, 128×128 grid, K_real = 5000, K_ctrl = 2000, R = 30, seed = 12345; torus arm uses 4 fixed geodesics [1,0],[0,1],[1,1],[2,3] and a 2000-sample simulated null. (The vendored R `torustest` was patched to run serially — its parallel `makeForkCluster` requires a socket module unavailable in this environment.)

Appendix A — kill-set samples: contributing structures and aggregated angle

Aggregated angle = circular mean (ϕ ± σϕ, ψ ± σψ) over the contributing structures (matches the pipeline centroid); σ is the circular standard deviation (Mardia), in degrees. The **position** is marked WIDE when σ > 15° on either axis. Per-structure angles are in `robustness_worst_angles_detail.csv`.

KDE-L1 arm

HELIX L-CTC:L-TTG

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
1	L-CTC	P30014:158	-127.4 ± 3.8	-6.7 ± 1.2	2	3V9W:A,159; 4KB1:A,159
2	L-CTC	P0AB91:152	-135.2	-16.9	1	1N8F:A,153
3	L-CTC	P37355:135	-84.2 ± 0.8	-14.1 ± 2.0	3	4MXD:A,136; 4MYD:A,136; 4MYS:A,136
4	L-CTC	P69506:12	-85.6	-13.9	1	5FNP:A,13
5	L-CTC	P0A9Q9:252	-81.0	-16.6	1	1T4B:A,253
6	L-CTC	P0A6R0:257	-77.6 ± 4.0	-3.1 ± 8.2	4	1EBL:A,258; 1HNJ:A,258; 6X7R:A,258; 6X7S:A,258
7	L-CTC	P33590:113	-77.2 ± 3.9	-15.5 ± 4.3	14	1ZLQ:A,92; 2NOO:A,92; 3MVW:A,92; 3MVX:A,92; 3MVZ:A,92; 3MZ9:A,92; 3MZB:A,92; 4I9D:A,92; 5L8D:A,92; 5MWU:A,92; 5ON1:A,92; 5ON5:A,92; 5ON8:A,92; 5ON9:A,92
8	L-CTC	P13035:428	-81.4	-18.8	1	2QCU:A,429

HELIX L-CTC:L-CTG

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
1	L-CTC	P0AB91:152	-135.2	-16.9	1	1N8F:A,153

TURN A-GCG:A-GCT

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
1	A-GCT	P27302:54	-59.2 ± 1.8	-33.4 ± 0.8	7	2R5N:A,55; 2R8O:A,55; 2R8P:A,55; 5HHT:A,55; 6RJC:A,55; 6TJ8:A,55; 6TJ9:A,55
2	A-GCT	Q6EZC2:192	-57.0	-33.5	1	2IY9:A,193
3	A-GCT	P76578:337	-58.3	-31.1	1	4ZJH:A,338
4	A-GCT	D7Y2H5:136	-57.9 ± 4.9	-37.8 ± 6.5	3	6P7O:A,137; 6P7P:A,137; 6P7Q:A,137
5	A-GCT	P08716:565	-144.0 ± 6.7	-46.3 ± 4.0	2	2FF7:A,566; 2PMK:A,566
6	A-GCT	P15005:58	-58.9 ± 3.8	-36.6 ± 3.4	3	6GCD:A,59; 6GCE:A,59; 6GCF:A,59
7	A-GCT	P0A6L2:48	-62.5 ± 0.7	-34.3 ± 0.1	2	2OJP:A,49; 2PUR:A,49
8	A-GCT	P0A6P9:83	-57.7	-28.5	1	2FYM:A,83

TURN P-CCC:P-CCG

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
1	P-CCC	Q68JC9:95	-56.2	+131.6	1	3EOI:A,75
2	P-CCC	P78067:186	-56.4	+132.8	1	2WLR:A,167
3	P-CCC	P09184:46	-55.7	+130.8	1	1VSR:A,47

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
4	P-CCC	P77489:351	-58.5	+134.3	1	5G5G:C,352
5	P-CCC	Q8VQD3:121	-55.1	+129.4	1	1YJ7:A,122
6	P-CCC	P00634:40	-55.9 ± 3.3	+129.0 ± 2.7	8	1ED8:A,19; 1ED9:A,19; 1Y6V:A,19; 1Y7A:A,19; 3BDF:A,19; 3BDG:A,19; 3BDG:B,19; 3TG0:A,19
7	P-CCC	P77366:95	-58.3	+137.3	1	4G9B:A,96
8	P-CCC	A0A1B3B7F6:173	-57.9	+137.2	1	6VNU:A,174

torus_p arm

HELIX L-CTC:L-TTG

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
1	L-CTC	P77454:302	-94.0	-3.5	1	1U60:A,303
2	L-CTC	P29208:237	-91.2 ± 1.5	-3.7 ± 0.5	2	1FHV:A,238; 1R6W:A,238
3	L-CTC	P21179:561	-103.9 ± 1.8	+0.4 ± 1.7	21	1CF9:A,562; 1P80:A,562; 3P9P:A,562; 3P9Q:A,562; 3PQ2:A,562; 3PQ3:A,562; 3PQ4:A,562; 3PQ5:A,562; 3PQ6:A,562; 3PQ7:A,562; 3PQ8:A,562; 3TTT:A,562; 3TTV:A,562; 3TTW:A,562; 3TTX:A,562; 4ENP:A,562; 4ENR:A,562; 4ENS:A,562; 4ENT:A,562; 4ENU:A,562; 4ENV:A,562
4	L-CTC	A0A400L9R1:36	-99.2 ± 3.8	+5.7 ± 3.9	2	6V2L:A,37; 6V2N:A,37
5	L-CTC	P15034:328	-87.9 ± 0.9	-5.8 ± 1.6	6	2BWS:A,328; 2BWV:A,328; 2BWX:A,328; 2V3X:A,328; 2V3Y:A,328; 2V3Z:A,328

HELIX L-CTC:L-CTG

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
1	L-CTC	P0AB91:152	-135.2	-16.9	1	1N8F:A,153
2	L-CTC	A0A400L9R1:36	-99.2 ± 3.8	+5.7 ± 3.9	2	6V2L:A,37; 6V2N:A,37
3	L-CTC	P18956:138	-124.2 ± 0.8	-32.4 ± 1.5	6	2DBW:A,139; 2DBX:A,139; 2DG5:A,139; 2Z8I:A,139; 2Z8K:A,139; 5B5T:A,139
4	L-CTC	P77454:302	-94.0	-3.5	1	1U60:A,303
5	L-CTC	P25740:248	-89.3 ± 1.3	-1.5 ± 1.1	2	2IV7:A,249; 2IW1:A,249
6	L-CTC	P29208:237	-91.2 ± 1.5	-3.7 ± 0.5	2	1FHV:A,238; 1R6W:A,238
7	L-CTC	P21179:561	-103.9 ± 1.8	+0.4 ± 1.7	21	1CF9:A,562; 1P80:A,562; 3P9P:A,562; 3P9Q:A,562; 3PQ2:A,562; 3PQ3:A,562; 3PQ4:A,562; 3PQ5:A,562; 3PQ6:A,562; 3PQ7:A,562; 3PQ8:A,562; 3TTT:A,562; 3TTV:A,562; 3TTW:A,562; 3TTX:A,562; 4ENP:A,562; 4ENR:A,562; 4ENS:A,562; 4ENT:A,562; 4ENU:A,562; 4ENV:A,562
8	L-CTC	P0A7J0:77	-87.1 ± 0.5	-1.2 ± 0.3	2	1G57:A,78; 1G58:A,78

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
9	L-CTC	P31133:163	-105.1 ± 2.3	+0.3 ± 1.5	5	4JDF:A,164; 6YE0:A,164; 6YE6:A,164; 6YE7:A,164; 6YE8:A,164
10	L-CTC	P15034:328	-87.9 ± 0.9	-5.8 ± 1.6	6	2BWS:A,328; 2BWV:A,328; 2BWX:A,328; 2V3X:A,328; 2V3Y:A,328; 2V3Z:A,328
11	L-CTC	P0A6L4:170	-82.4 ± 1.7	-7.0 ± 1.9	4	2WKJ:A,171; 3LBM:A,171; 3LCG:A,171; 4UUI:A,171
12	L-CTC	Q9S5G5:161	-84.3 ± 1.6	-1.7 ± 1.8	4	2FPR:A,163; 2FPU:A,163; 2FPW:A,163; 2FPX:A,163
13	L-CTC	A0A0F3U9S7:506	-79.3 ± 0.5	-10.2 ± 0.8	4	6SPN:A,506; 6SPO:A,506; 6SPP:A,506; 6SPQ:A,506
14	L-CTC	P41409:22	-78.3	-11.1	1	1YOE:A,23
15	L-CTC	P24182:443	-78.2	-11.9	1	2W70:A,444
16	L-CTC	P69441:212	-80.0	-12.4	1	4X8L:A,213
17	L-CTC	A0MMD0:112	-76.8	-10.2	1	1QTX:A,112
18	L-CTC	P0AAI9:306	-73.2	-16.9	1	2G2O:A,307
19	L-CTC	P0A881:74	-111.6 ± 8.1	-4.1 ± 6.3	10	1JHG:A,75; 3SSW:N,75; 6EKP:A,75; 6ELB:A,75; 6ELG:A,75; 6ENI:A,75; 6ENN:A,75; 6F7G:A,75; 6F9K:A,75; 6FAL:A,75
20	L-CTC	P33590:113	-77.2 ± 3.9	-15.5 ± 4.3	14	1ZLQ:A,92; 2NOO:A,92; 3MVW:A,92; 3MVX:A,92; 3MVZ:A,92; 3MZ9:A,92; 3MZB:A,92; 4I9D:A,92; 5L8D:A,92; 5MWU:A,92; 5ON1:A,92; 5ON5:A,92; 5ON8:A,92; 5ON9:A,92

HELIX L-CTC:L-CTT

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
1	L-CTC	Q9S5G5:161	-84.3 ± 1.6	-1.7 ± 1.8	4	2FPR:A,163; 2FPU:A,163; 2FPW:A,163; 2FPX:A,163
2	L-CTC	A0MMD0:112	-76.8	-10.2	1	1QTX:A,112
3	L-CTC	P0A7J0:77	-87.1 ± 0.5	-1.2 ± 0.3	2	1G57:A,78; 1G58:A,78
4	L-CTC	P0A6L4:170	-82.4 ± 1.7	-7.0 ± 1.9	4	2WKJ:A,171; 3LBM:A,171; 3LCG:A,171; 4UUI:A,171
5	L-CTC	P25740:248	-89.3 ± 1.3	-1.5 ± 1.1	2	2IV7:A,249; 2IW1:A,249
6	L-CTC	A0A0F3U9S7:506	-79.3 ± 0.5	-10.2 ± 0.8	4	6SPN:A,506; 6SPO:A,506; 6SPP:A,506; 6SPQ:A,506
7	L-CTC	P41409:22	-78.3	-11.1	1	1YOE:A,23
8	L-CTC	P24182:443	-78.2	-11.9	1	2W70:A,444
9	L-CTC	A0A400L9R1:36	-99.2 ± 3.8	+5.7 ± 3.9	2	6V2L:A,37; 6V2N:A,37
10	L-CTC	P69441:212	-80.0	-12.4	1	4X8L:A,213
11	L-CTC	P29208:237	-91.2 ± 1.5	-3.7 ± 0.5	2	1FHV:A,238; 1R6W:A,238
12	L-CTC	P15034:328	-87.9 ± 0.9	-5.8 ± 1.6	6	2BWS:A,328; 2BWV:A,328; 2BWX:A,328; 2V3X:A,328; 2V3Y:A,328; 2V3Z:A,328
13	L-CTC	P0AB91:152	-135.2	-16.9	1	1N8F:A,153
14	L-CTC	Q9EVJ6:109	-65.9	-18.9	1	3FRQ:A,110
15	L-CTC	P0AAI9:306	-73.2	-16.9	1	2G2O:A,307
16	L-CTC	P77454:302	-94.0	-3.5	1	1U60:A,303

#	codon	position	φ ± σ (°)	ψ ± σ (°)	n	structures
17	L-CTC	P33590:113	-77.2 ± 3.9	-15.5 ± 4.3	14	1ZLQ:A,92; 2NOO:A,92; 3MVW:A,92; 3MVX:A,92; 3MVZ:A,92; 3MZ9:A,92; 3MZB:A,92; 4I9D:A,92; 5L8D:A,92; 5MWU:A,92; 5ON1:A,92; 5ON5:A,92; 5ON8:A,92; 5ON9:A,92
18	L-CTC	P0A7I7:138	-69.2	-19.3	1	2BZ1:A,139
19	L-CTC	P75691:250	-63.0	-21.7	1	1UUF:A,251
20	L-CTC	P31808:139	-67.3	-20.8	1	3F1L:A,140

HELIX R-AGG:R-CGA

#	codon	position	φ ± σ (°)	ψ ± σ (°)	n	structures
1	R-AGG	Q93ET9:278	-51.8	-33.8	1	4DCA:A,279
2	R-CGA	P0A7D1:186	-70.3	-48.5	1	2PTH:A,186
3	R-AGG	Q8X834:258	-55.2	-32.2	1	4LGJ:A,250
4	R-AGG	Q7DI68:193	-59.3	-27.4	1	4M1U:A,172
5	R-CGA	Q93ET8:123	-60.3	-54.4	1	6BFF:A,124
6	R-CGA	P0A6E4:31	-69.5	-46.2	1	1K92:A,31
7	R-CGA	P06721:301	-79.0 ± 0.4	-39.3 ± 2.6	4	2FQ6:A,302; 2GQN:A,302; 4ITG:A,302; 4ITX:A,302
8	R-AGG	P77366:70	-62.5	-37.2	1	4G9B:A,71
9	R-CGA	A0A0H2Z2W8:278	-74.6	-40.6	1	6MGC:A,279
10	R-AGG	P33362:254	-58.9	-31.1	1	4WEP:A,255
11	R-AGG	P03007:134	-63.0	-36.5	1	2GUI:A,135
12	R-CGA	P00557:291	-67.5 ± 2.0	-45.8 ± 2.2	3	3W0O:A,292; 3W0Q:A,292; 3W0S:A,292

TURN A-GCG:A-GCT

#	codon	position	φ ± σ (°)	ψ ± σ (°)	n	structures
1	A-GCG	P33590:421 **WIDE**	-65.9 ± 26.2	+149.1 ± 16.8	14	1ZLQ:A,400; 2NOO:A,400; 3MVW:A,400; 3MVX:A,400; 3MVZ:A,400; 3MZ9:A,400; 3MZB:A,400; 4I9D:A,400; 5L8D:A,400; 5MWU:A,400; 5ON1:A,400; 5ON5:A,400; 5ON8:A,400; 5ON9:A,400
2	A-GCG	P77366:191	-64.0	+148.1	1	4G9B:A,192
3	A-GCG	P08191:254	-61.1	+146.5	1	4XO9:A,234
4	A-GCG	Q8X6R9:67	-68.2 ± 3.1	+152.4 ± 3.3	3	3ZDA:A,68; 3ZDB:A,68; 3ZDC:A,68
5	A-GCG	Q8FBQ3:110	-60.0	+146.6	1	2FT0:A,111

Appendix B — kill-set samples: contributing structures and aggregated angle (outlier-cleaned aggregation)

Aggregated angle = circular mean (ϕ ± σϕ, ψ ± σψ) over the contributing structures (outlier structures removed); σ is the circular standard deviation (Mardia), in degrees. The **position** is marked WIDE when σ > 15° on either axis. Per-structure angles are in `robustness_worst_angles_detail.csv`.

KDE-L1 arm

HELIX L-CTC:L-TTG

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
1	L-CTC	P30014:158	-127.4 ± 3.8	-6.7 ± 1.2	2	3V9W:A,159; 4KB1:A,159
2	L-CTC	P0AB91:152	-135.2	-16.9	1	1N8F:A,153
3	L-CTC	P37355:135	-84.2 ± 0.8	-14.1 ± 2.0	3	4MXD:A,136; 4MYD:A,136; 4MYS:A,136
4	L-CTC	P69506:12	-85.6	-13.9	1	5FNP:A,13
5	L-CTC	P0A9Q9:252	-81.0	-16.6	1	1T4B:A,253
6	L-CTC	P0A6R0:257	-77.6 ± 4.0	-3.1 ± 8.2	4	1EBL:A,258; 1HNJ:A,258; 6X7R:A,258; 6X7S:A,258
7	L-CTC	P33590:113	-77.2 ± 3.9	-15.5 ± 4.3	14	1ZLQ:A,92; 2NOO:A,92; 3MVW:A,92; 3MVX:A,92; 3MVZ:A,92; 3MZ9:A,92; 3MZB:A,92; 4I9D:A,92; 5L8D:A,92; 5MWU:A,92; 5ON1:A,92; 5ON5:A,92; 5ON8:A,92; 5ON9:A,92
8	L-CTC	P13035:428	-81.4	-18.8	1	2QCU:A,429

HELIX L-CTC:L-CTG

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
1	L-CTC	P0AB91:152	-135.2	-16.9	1	1N8F:A,153

TURN A-GCG:A-GCT

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
1	A-GCT	P27302:54	-59.2 ± 1.8	-33.4 ± 0.8	7	2R5N:A,55; 2R8O:A,55; 2R8P:A,55; 5HHT:A,55; 6RJC:A,55; 6TJ8:A,55; 6TJ9:A,55
2	A-GCT	Q6EZC2:192	-57.0	-33.5	1	2IY9:A,193
3	A-GCT	P76578:337	-58.3	-31.1	1	4ZJH:A,338
4	A-GCT	P08716:565	-144.0 ± 6.7	-46.3 ± 4.0	2	2FF7:A,566; 2PMK:A,566
5	A-GCT	D7Y2H5:136	-57.9 ± 4.9	-37.8 ± 6.5	3	6P7O:A,137; 6P7P:A,137; 6P7Q:A,137
6	A-GCT	P15005:58	-58.9 ± 3.8	-36.6 ± 3.4	3	6GCD:A,59; 6GCE:A,59; 6GCF:A,59
7	A-GCT	P0A6P9:83	-57.7	-28.5	1	2FYM:A,83
8	A-GCT	P0A6L2:48	-62.5 ± 0.7	-34.3 ± 0.1	2	2OJP:A,49; 2PUR:A,49

TURN P-CCC:P-CCG

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
1	P-CCC	Q68JC9:95	-56.2	+131.6	1	3EOI:A,75
2	P-CCC	P78067:186	-56.4	+132.8	1	2WLR:A,167

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
3	P-CCC	P09184:46	-55.7	+130.8	1	1VSR:A,47
4	P-CCC	Q8VQD3:121	-55.1	+129.4	1	1YJ7:A,122
5	P-CCC	P77489:351	-58.5	+134.3	1	5G5G:C,352
6	P-CCC	P00634:40	-55.9 ± 3.3	+129.0 ± 2.7	8	1ED8:A,19; 1ED9:A,19; 1Y6V:A,19; 1Y7A:A,19; 3BDF:A,19; 3BDG:A,19; 3BDG:B,19; 3TG0:A,19
7	P-CCC	P77366:95	-58.3	+137.3	1	4G9B:A,96
8	P-CCC	A0A1B3B7F6:173	-57.9	+137.2	1	6VNU:A,174

torus_p arm

HELIX L-CTC:L-TTG

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
1	L-CTC	P77454:302	-94.0	-3.5	1	1U60:A,303
2	L-CTC	P29208:237	-91.2 ± 1.5	-3.7 ± 0.5	2	1FHV:A,238; 1R6W:A,238
3	L-CTC	P21179:561	-103.9 ± 1.8	+0.4 ± 1.7	21	1CF9:A,562; 1P80:A,562; 3P9P:A,562; 3P9Q:A,562; 3PQ2:A,562; 3PQ3:A,562; 3PQ4:A,562; 3PQ5:A,562; 3PQ6:A,562; 3PQ7:A,562; 3PQ8:A,562; 3TTT:A,562; 3TTV:A,562; 3TTW:A,562; 3TTX:A,562; 4ENP:A,562; 4ENR:A,562; 4ENS:A,562; 4ENT:A,562; 4ENU:A,562; 4ENV:A,562
4	L-CTC	A0A400L9R1:36	-99.2 ± 3.8	+5.7 ± 3.9	2	6V2L:A,37; 6V2N:A,37
5	L-CTC	P15034:328	-87.9 ± 0.9	-5.8 ± 1.6	6	2BWS:A,328; 2BWV:A,328; 2BWX:A,328; 2V3X:A,328; 2V3Y:A,328; 2V3Z:A,328

HELIX L-CTC:L-CTG

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
1	L-CTC	P0AB91:152	-135.2	-16.9	1	1N8F:A,153
2	L-CTC	A0A400L9R1:36	-99.2 ± 3.8	+5.7 ± 3.9	2	6V2L:A,37; 6V2N:A,37
3	L-CTC	P18956:138	-124.2 ± 0.8	-32.4 ± 1.5	6	2DBW:A,139; 2DBX:A,139; 2DG5:A,139; 2Z8I:A,139; 2Z8K:A,139; 5B5T:A,139
4	L-CTC	P77454:302	-94.0	-3.5	1	1U60:A,303
5	L-CTC	P25740:248	-89.3 ± 1.3	-1.5 ± 1.1	2	2IV7:A,249; 2IW1:A,249
6	L-CTC	P29208:237	-91.2 ± 1.5	-3.7 ± 0.5	2	1FHV:A,238; 1R6W:A,238
7	L-CTC	P21179:561	-103.9 ± 1.8	+0.4 ± 1.7	21	1CF9:A,562; 1P80:A,562; 3P9P:A,562; 3P9Q:A,562; 3PQ2:A,562; 3PQ3:A,562; 3PQ4:A,562; 3PQ5:A,562; 3PQ6:A,562; 3PQ7:A,562; 3PQ8:A,562; 3TTT:A,562; 3TTV:A,562; 3TTW:A,562; 3TTX:A,562; 4ENP:A,562; 4ENR:A,562; 4ENS:A,562; 4ENT:A,562; 4ENU:A,562; 4ENV:A,562

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
8	L-CTC	P0A7J0:77	-87.1 ± 0.5	-1.2 ± 0.3	2	1G57:A,78; 1G58:A,78
9	L-CTC	P31133:163	-105.1 ± 2.3	+0.3 ± 1.5	5	4JDF:A,164; 6YE0:A,164; 6YE6:A,164; 6YE7:A,164; 6YE8:A,164
10	L-CTC	P15034:328	-87.9 ± 0.9	-5.8 ± 1.6	6	2BWS:A,328; 2BWV:A,328; 2BWX:A,328; 2V3X:A,328; 2V3Y:A,328; 2V3Z:A,328
11	L-CTC	P0A6L4:170	-82.4 ± 1.7	-7.0 ± 1.9	4	2WKJ:A,171; 3LBM:A,171; 3LCG:A,171; 4UUI:A,171
12	L-CTC	Q9S5G5:161	-84.3 ± 1.6	-1.7 ± 1.8	4	2FPR:A,163; 2FPU:A,163; 2FPW:A,163; 2FPX:A,163
13	L-CTC	A0A0F3U9S7:506	-79.3 ± 0.5	-10.2 ± 0.8	4	6SPN:A,506; 6SPO:A,506; 6SPP:A,506; 6SPQ:A,506
14	L-CTC	P41409:22	-78.3	-11.1	1	1YOE:A,23
15	L-CTC	P24182:443	-78.2	-11.9	1	2W70:A,444
16	L-CTC	P69441:212	-80.0	-12.4	1	4X8L:A,213
17	L-CTC	A0MMD0:112	-76.8	-10.2	1	1QTX:A,112
18	L-CTC	P0AAI9:306	-73.2	-16.9	1	2G2O:A,307
19	L-CTC	P0A881:74	-111.6 ± 8.1	-4.1 ± 6.3	10	1JHG:A,75; 3SSW:N,75; 6EKP:A,75; 6ELB:A,75; 6ELG:A,75; 6ENI:A,75; 6ENN:A,75; 6F7G:A,75; 6F9K:A,75; 6FAL:A,75
20	L-CTC	P33590:113	-77.2 ± 3.9	-15.5 ± 4.3	14	1ZLQ:A,92; 2NOO:A,92; 3MVW:A,92; 3MVX:A,92; 3MVZ:A,92; 3MZ9:A,92; 3MZB:A,92; 4I9D:A,92; 5L8D:A,92; 5MWU:A,92; 5ON1:A,92; 5ON5:A,92; 5ON8:A,92; 5ON9:A,92

HELIX L-CTC:L-CTT

#	codon	position	ϕ ± σ (°)	ψ ± σ (°)	n	structures
1	L-CTC	Q9S5G5:161	-84.3 ± 1.6	-1.7 ± 1.8	4	2FPR:A,163; 2FPU:A,163; 2FPW:A,163; 2FPX:A,163
2	L-CTC	A0MMD0:112	-76.8	-10.2	1	1QTX:A,112
3	L-CTC	P0A7J0:77	-87.1 ± 0.5	-1.2 ± 0.3	2	1G57:A,78; 1G58:A,78
4	L-CTC	P0A6L4:170	-82.4 ± 1.7	-7.0 ± 1.9	4	2WKJ:A,171; 3LBM:A,171; 3LCG:A,171; 4UUI:A,171
5	L-CTC	P25740:248	-89.3 ± 1.3	-1.5 ± 1.1	2	2IV7:A,249; 2IW1:A,249
6	L-CTC	A0A0F3U9S7:506	-79.3 ± 0.5	-10.2 ± 0.8	4	6SPN:A,506; 6SPO:A,506; 6SPP:A,506; 6SPQ:A,506
7	L-CTC	P41409:22	-78.3	-11.1	1	1YOE:A,23
8	L-CTC	P24182:443	-78.2	-11.9	1	2W70:A,444
9	L-CTC	A0A400L9R1:36	-99.2 ± 3.8	+5.7 ± 3.9	2	6V2L:A,37; 6V2N:A,37
10	L-CTC	P69441:212	-80.0	-12.4	1	4X8L:A,213
11	L-CTC	P29208:237	-91.2 ± 1.5	-3.7 ± 0.5	2	1FHV:A,238; 1R6W:A,238
12	L-CTC	P15034:328	-87.9 ± 0.9	-5.8 ± 1.6	6	2BWS:A,328; 2BWV:A,328; 2BWX:A,328; 2V3X:A,328; 2V3Y:A,328; 2V3Z:A,328
13	L-CTC	P0AB91:152	-135.2	-16.9	1	1N8F:A,153
14	L-CTC	Q9EVJ6:109	-65.9	-18.9	1	3FRQ:A,110
15	L-CTC	P0AAI9:306	-73.2	-16.9	1	2G2O:A,307
16	L-CTC	P77454:302	-94.0	-3.5	1	1U60:A,303

#	codon	position	φ ± σ (°)	ψ ± σ (°)	n	structures
17	L-CTC	P33590:113	-77.2 ± 3.9	-15.5 ± 4.3	14	1ZLQ:A,92; 2NOO:A,92; 3MVW:A,92; 3MVX:A,92; 3MVZ:A,92; 3MZ9:A,92; 3MZB:A,92; 4I9D:A,92; 5L8D:A,92; 5MWU:A,92; 5ON1:A,92; 5ON5:A,92; 5ON8:A,92; 5ON9:A,92
18	L-CTC	P0A7I7:138	-69.2	-19.3	1	2BZ1:A,139
19	L-CTC	P75691:250	-63.0	-21.7	1	1UUF:A,251
20	L-CTC	P31808:139	-67.3	-20.8	1	3F1L:A,140

HELIX R-AGG:R-CGA

#	codon	position	φ ± σ (°)	ψ ± σ (°)	n	structures
1	R-AGG	Q93ET9:278	-51.8	-33.8	1	4DCA:A,279
2	R-CGA	P0A7D1:186	-70.3	-48.5	1	2PTH:A,186
3	R-AGG	Q8X834:258	-55.2	-32.2	1	4LGJ:A,250
4	R-AGG	Q7DI68:193	-59.3	-27.4	1	4M1U:A,172
5	R-CGA	Q93ET8:123	-60.3	-54.4	1	6BFF:A,124
6	R-CGA	P0A6E4:31	-69.5	-46.2	1	1K92:A,31
7	R-CGA	P06721:301	-79.0 ± 0.4	-39.3 ± 2.6	4	2FQ6:A,302; 2GQN:A,302; 4ITG:A,302; 4ITX:A,302
8	R-AGG	P77366:70	-62.5	-37.2	1	4G9B:A,71
9	R-CGA	A0A0H2Z2W8:278	-74.6	-40.6	1	6MGC:A,279
10	R-AGG	P33362:254	-58.9	-31.1	1	4WEP:A,255
11	R-AGG	P03007:134	-63.0	-36.5	1	2GUI:A,135
12	R-CGA	P00557:291	-67.5 ± 2.0	-45.8 ± 2.2	3	3W0O:A,292; 3W0Q:A,292; 3W0S:A,292

TURN A-GCG:A-GCT

#	codon	position	φ ± σ (°)	ψ ± σ (°)	n	structures
1	A-GCG	P77366:191	-64.0	+148.1	1	4G9B:A,192
2	A-GCG	Q8FBQ3:110	-60.0	+146.6	1	2FT0:A,111
3	A-GCG	Q8X6R9:67	-68.2 ± 3.1	+152.4 ± 3.3	3	3ZDA:A,68; 3ZDB:A,68; 3ZDC:A,68
4	A-GCG	P08191:254	-61.1	+146.5	1	4XO9:A,234
5	A-GCG	Q46925:264	-62.8	+149.5	1	4LW2:A,265